MILITARY SPECIFICATION

LENS, REFLECTING SIGHT, ASSEMBLY

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments

1. CLASSIFICATION

1.1 Type.—Lens, reflecting sight, assembly, shall be of only one type. A different kind of reticle, however, is required for each ballistic application.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications.—The following specifications and specifications referenced on drawings in 2.2, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATION

UU-T-111—Tape; Paper, Gummed (Kraft).

MILITARY SPECIFICATIONS

JAN-P-100—Packaging and Packing for Overseas Shipment—General Specification.

JAN-P-106—Packaging and Packing for Overseas Shipment.—Boxes; Wood, Nailed.

JAN-P-108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

JAN-P-116—Packaging and Packing for Overseas Shipment—Preservation, Methods of.

JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

AIR FORCE-NAVY AERONAUTICAL SPECIFICATION

AN-W-18—Wadding; Creped Cellulose.

U. S. ARMY SPECIFICATIONS

51-7—Instruments, Fire-Control, General Specification Covering the Manufacture and Inspection of.

51-70-2—Optical Components for Fire Control Instruments, General Specification Governing the Manufacture, Assembly, and Inspection of.

100-2—Standard Specification for Marking Shipments by Contractors.

(Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Drawings.—The following drawings and drawings referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification:

ARMY ORDNANCE DEPARTMENT DRAWINGS

B7161561—Reticle.

B7161563—Lens, Reflecting Sight, Assembly (For Launcher, Rocket, 2.36-inch, M9A1 and M18).

C7162889—Lens, Reflecting Sight, Assembly (For Launcher, Rocket, 3.5-inch, M20).

C7677030—Reticle.

(Copies of drawings should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Unless otherwise specified, lens, reflecting sight, assembly, shall conform to specifications 51-7 and 51-70-2.

3.2 Material.—Material shall conform to the applicable specifications and drawings.

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3.3 Reticle focus.—The reticle shall be in sharp focus between zero and minus 0.5 diopter.

3.4 Lens power.—Lens power shall be between 0 and minus 0.2 diopter.

3.5 Astigmatism.—There shall be no apparent astigmatism at the center of the field.

3.6 Deviation.—The included angle of deviation of the lens assembly shall not exceed 1°.

3.7 Light transmission.—The light transmission of the lens assembly shall be between 30 and 50 percent.

3.8 Angular subtense.—The reticle pattern shall conform to the appropriate reticle drawing.

3.8.1 With reticle shown by drawing C7677030 in the lens assembly for the 3.5-inch rocket launcher, the following values shall be obtained:

3.8.1.1 The horizontal angular subtense from the central vertical line to the center of the inner short vertical line shall be 44 ±2 mils.

3.8.1.2 The horizontal angular subtense from the central vertical line to the mid point intersection of the ellipse with the horizontal axis shall be 89 ±3 mils.

3.8.1.3 The horizontal angular subtense from the central vertical line to the center of the outer short vertical line shall be 134 ±4 mils.

3.8.2 With reticle shown by drawing B7161561 in the lens assembly for the 2.36-inch rocket launcher, the following values shall be obtained:

3.8.2.1 The angular subtense of the mean diameter of circles A, B, C, and D shall measure, respectively, 107 ±6 mils, 212 ±7 mils, 317 ±8 mils, and 420 ±9 mils.

3.9 Waterproofing.—The lens assembly shall not permit entry of moisture into its interior when immersed in water.

3.10 Temperature.—The temperature requirements specified in specification 51–7 for type II instruments shall apply.

3.11 Workmanship.—Workmanship shall conform to specifications 51–7 and 51–70–2. The complete assembly shall be free of scratches, digs, inclusions, and other defects; and there shall be no particles of dirt, flakes of reflecting material, or foreign matter inside the sight assembly after being subjected to the vibration test.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—

4.1.1 *Lot.*—A lot shall consist of all assemblies or parts in a single shipment.

4.1.2 *Sampling procedure.*—Unless otherwise specified, each lens assembly and extra part shall be inspected for conformance with applicable specifications and drawings, and they shall be subject to inspection and test during process of manufacture and assembly as provided in specifications 51–7 and 51–70–2.

4.2 Inspection.—

4.2.1 Unless otherwise specified, inspection shall be at the point of manufacture.

4.2.2 *Visual.*—Examine the lens assembly for visual defects 20 inches from an opal glass screen illuminated by a frosted 25-watt electric bulb placed 5 inches behind the glass.

4.3 Tests.—

4.3.1 *Deviation.*—Rotate the lens assembly 360° in a "V" block and note the maximum deviation of the optical axis.

4.3.2 *Reticle focus, lens power, astigmatism, and light transmission.*—The reticle focus, lens power, astigmatism, and light transmission shall be determined by conventional methods.

4.3.3 *Angular subtense.*—Either a collimator or a target situated at least 80 feet from the lens assembly may be used to determine angular subtense. The observer's eye should be approximately 1 inch from the rear surface of the reticle.

4.3.4 *Waterproofing.*—Lens assembly shall be completely immersed in water to a pressure of 2.6 p. s. i. (6 feet of water) for 30 minutes. After removal, the lens assembly shall be dried and allowed to stand for 24 hours and then visually examined for evidence of internal moisture.

4.3.5 *Temperature.*—The temperature test shall be made as specified in specification 51–7 for type II instruments.

* 5. PREPARATION FOR DELIVERY

5.1 Preservation.—The reflecting sight lens assembly shall be cleaned in accordance with the applicable requirements of specification JAN–P–116. Wrap each lens assembly with neutral tissue paper and overwrap with cellulose

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wadding as specified for type I of specification AN–W–18.

5.2 Packaging.—

5.2.1 *Unit packaging.*—Each reflecting sight lens assembly shall be unit packaged in accordance with specification JAN–P–116, method II (use type I moisture vaporproof barrier material and type V, grade A desiccant).

5.2.2 *Intermediate containers.*—Twenty-four unit packaged lenses shall be packaged in an intermediate container conforming to type SF or CF, grade 5 or 6, any style, of specification JAN–P–108. Apply gummed tape over the seams. Tape shall conform to grade B of specification UU–T–111. Each intermediate pack shall be overwrapped with waterproof barrier material conforming to specification JAN–P–125.

5.2.3 Immobilize the contents of each pack. Except as indicated, packaging materials shall conform to the general requirements of specification JAN–P–100.

5.3 Packing.—

5.3.1 *For domestic shipment.*—Unless otherwise specified, reflecting sight lens assemblies shall be packed in substantial commercial wood containers of the type, size, and kind commonly used for the purpose in such a manner as to insure acceptance by common carriers, for safe transportation, at the lowest rate, to point of delivery.

5.3.2 *For overseas shipment.*—Reflecting sight lens assemblies, preserved and packaged as specified in 5.1 and 5.2, shall be packed in shipping containers conforming to the applicable requirements of specification JAN–P–106. Each box shall contain a waterproof bag or case liner (in which the contents are packed) fabricated from material conforming to the applicable requirements of specification JAN–P–125. Strap the box in accordance with the applicable box specification.

5.3.3 The contents of both domestic and overseas shipping containers shall be immobilized.

5.4 Marking for shipment.—

5.4.1 *Interior containers.*—Unit packs, intermediate packs, and waterproof overwraps shall have the following information printed or marked in a legible manner either directly on the containers (unit and intermediate) and overwraps or on labels attached to the containers and overwraps:

Quantity _____
Nomenclature _____
Ord dwg number _____
Stock number _____
Mfd by _____
Date packed: Month _____
Year _____

5.4.2 *Containers.*—In addition to any special markings required by the contract or order, shipments shall be marked in accordance with specification 100–2.

6. NOTES

6.1 Intended use.—Lens, reflecting sight, assembly, are intended for use as sighting devices for the 2.36-inch and the 3.5-inch rocket launchers.

6.2 Ordering data.—Invitation for bids and contracts should specify the following:

a. Title, number, and date of this specification.

b. Kind of reticle required. (See 1.1.)

c. Domestic or overseas packing required. (See 5.3.)

Notice.—This specification together with specifications and drawings pertaining to it and bearing a "NOTICE" of similar restrictions is intended for use in connection with procurement by the United States Government, and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any purpose except when specifically authorized by the Chief of Ordnance.

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